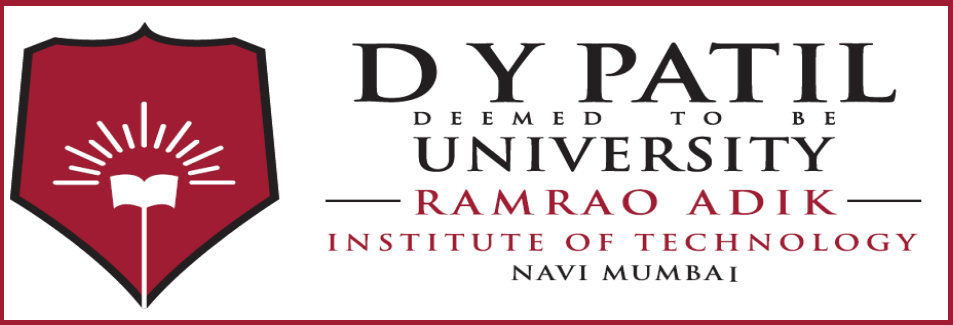


Embedded System
(Multidisciplinary Minor)
Module 1: Overview of Digital System

Index

Lecture 1 – Number System: Introduction to Binary, Octal, Decimal, and Hexadecimal number systems

Lecture 2 – conversion methods

Lecture 3 – Logic Gates: Representation of basic logic gates with symbols and truth tables.

Lecture 4 –Circuit diagrams and working principles of Adders and Multiplexers, Demultiplexers

Lecture 5– R-S Flip-Flops, and D Flip-Flops.

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Number System



Introduction to Digital Logic Circuits

- **Digital Electronics represents information , with only two discrete values.(0,1).**

- **Ideally**

  “no voltage” (e.g., 0v) represents a 0
  “full source voltage” (e.g., 5v) represents a 1

- **Realistically**

  “low voltage” (e.g., <1v) represents a 0
  “high voltage” (e.g., >4v) represents a 1

- **We achieve these discrete values by using switches.**

- **We use transistor switches, which operates at high speed, electronically, a small in size.**

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Analog versus Digital

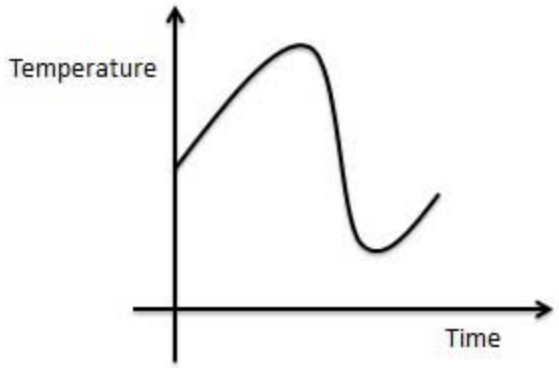
- **Analog** systems process time-varying signals that can take on any value across a continuous range of voltages (in electrical/electronics systems).

- **Digital** systems process time-varying signals that can take on only one of two discrete values of voltages (in electrical/electronics systems).

➢ Discrete values are called 1 and 0
(ON and OFF, HIGH and LOW, TRUE and FALSE, etc.)

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Graphical representation of Analog Signal



➢Disadvantage of Analog Systems:

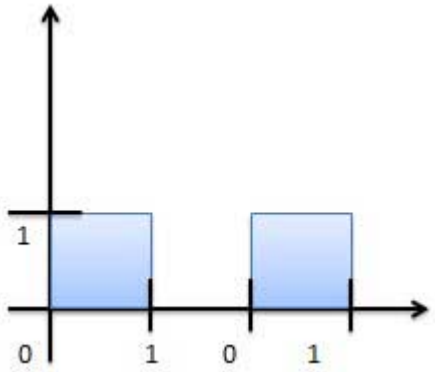
- Less accuracy
- Less versatility
- More noise effect
- More distortion

➢Examples of the Analog System :

- Filter
- Amplifiers
- Television receiver
- Motor speed controller

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Graphical Representation of Digital Signal



➢Advantage of Digital Systems

- More accuracy
- More versatility
- Less distortion
- Easy communication
- Possible storage of information

➢Examples of the digital systems are :

- Registers
- Flip-flop
- Counters
- Microprocessors

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Number System

- Many number systems are in use in digital technology.

- The most common are :
  - Decimal (Base 10)
  - Binary (Base 2)
  - Octal (Base 8)
  - Hexadecimal (Base 16)

- The decimal system is the number system that we use everyday.

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Number System

- A number system is defined as a system of writing for expressing numbers.

- It is the mathematical notation for representing numbers by using digits or other symbols in a consistent manner.

- It provides a unique representation of every number

- The value of any digit in a number can be determined by:
  - The digit
  - Its position in the number
  - The base of the number system

- In general, if *b* is the base, a number in the numeral system of base *b* can be expressed in the form:

$$a_n b^n + a_{n-1} b^{n-1} + a_{n-2} b^{n-2} + \ldots + a_0 b^0$$

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Number System

Common Number Systems

System	Base	Symbols
Decimal	10	0,1,2. . .. 9
Binary	2	0,1
Octal	8	0,1,. . .. 7
Hexa-Decimal	16	0,1,. . ..9 ,A,B,. . ...F

•When we represent numbers using 1s and 0s, we're using the **binary number system**.

•This system is fundamental to everything in digital electronics

•First, we'll briefly review the **decimal number system** that you've used for most of your life.

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Number System

•The position of each digit in a weighted number system is assigned a weight based on the **base** or **radix** of the system.

Decimal Numbers

•The **base of decimal numbers is ten**, as only ten symbols (0 through 9) are used to represent any number.

•The column weights of decimal numbers are powers of ten that increase from right to left beginning with $10^0 = 1$: $\ldots 10^5\ 10^4\ 10^3\ 10^2\ 10^1\ 10^0 .\ 10^{-1}\ 10^{-2} \ldots$

•Decimal numbers can be expressed as the sum of the products of each digit times the column value for that digit. Thus, the number 9240 can be expressed as: $(9 \times 10^3) + (2 \times 10^2) + (4 \times 10^1) + (0 \times 10^0)$

or

$9 \times 1{,}000 + 2 \times 100 + 4 \times 10 + 0 \times 1$

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Binary Number System

Binary Numbers

•Binary has **a base of two** and uses the digits 0 and 1 to represent quantities.

•The column weights of binary numbers are powers of two that increase from right to left beginning with $2^0 = 1$: $\ldots 2^5\ 2^4\ 2^3\ 2^2\ 2^1\ 2^0 .\ 2^{-1}\ 2^{-2}\ 2^{-3} \ldots$

Octal Numbers

•Octal numbers uses **eight** characters to represent numbers: the numbers 0 to 7.

•The column weights of octal numbers are power of 7 that increases from right to left beginning with $8^0 = 1$ $\ldots\ldots 8^4\ 8^3\ 8^2\ 8^1\ 8^0\ .8^{-1}\ 8^{-2} 8^{-3} \ldots\ldots$

•Binary number can easily be converted to octal by grouping bits 3 at a time and writing the equivalent octal character for each group.

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Hexadecimal Number System

Hexadecimal Numbers

•Hexadecimal uses **sixteen** characters to represent numbers: the numbers 0 through 9 and the alphabetic characters A through F.

•Large binary numbers can easily be converted to hexadecimal by grouping bits 4 at a time and writing the equivalent hex character.

•Hexadecimal is a weighted number system. The column weights are powers of 16, which increase from right to left.

$\ldots\ldots 16^4\ 16^3\ 16^2\ 16^1\ 16^0\ .16^{-1}\ 16^{-2} \qquad 16^{-3}\ \ldots\ldots$

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Number System Table

Decimal	Octal	Hexa	Binary	BCD
0	0	0	0000	0000
1	1	1	0001	0001
2	2	2	0010	0010
3	3	3	0011	0011
4	4	4	0100	0100
5	5	5	0101	0101
6	6	6	0110	0110
7	7	7	0111	0111
8	10	8	1000	1000
9	11	9	1001	1001
10	12	A	1010	0001 0000
11	13	B	1011	0001 0001
12	14	C	1100	0001 0010
13	15	D	1101	0001 0011
14	16	E	1110	0001 0100
15	17	F	1111	0001 0101

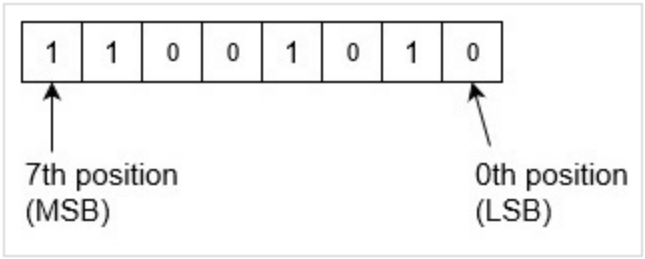
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Number System Conversion



Number System Conversion

- Convert binary number 11001010 into decimal number

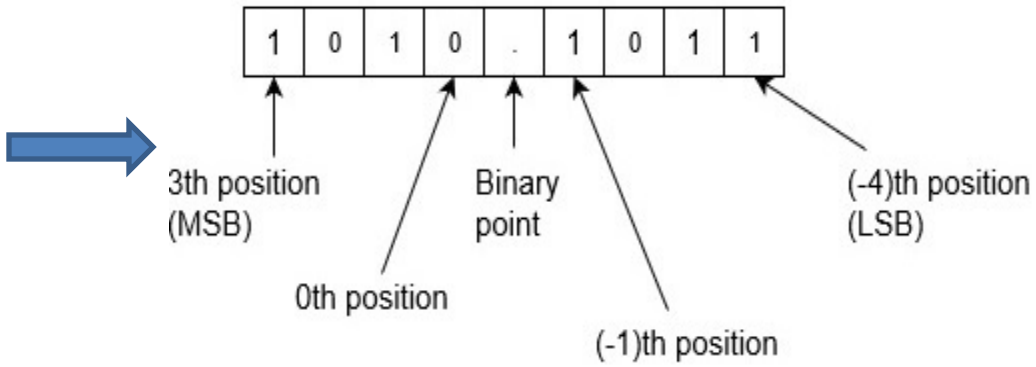


$$= (11001010)_2$$
$$= 1x2^7+1x2^6+0x2^5+0x2^4+1x2^3+0x2^2+1x2^1+0x2^0$$
$$= 128+64+0+0+8+0+2+0$$
$$= (202)_{10}$$

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Number System Conversion

➢Convert binary number 1010.1011 into decimal number. Since there is a binary point here with fractional part.



$$= (1010.1011)_2$$

$$= 1x2^3+0x2^2+1x2^1+0x2^0+1x2^{-1}+0x2^{-2}+1x2^{-3}+1x2^{-4}$$

$$= 8+0+2+0+0.5+0+0.125+0.0625$$

$$= (10.6875)_{10}$$

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Number System Conversion

- Convert decimal number 112 into binary number.

Division	Remainder (R)
112 / 2 = 56	0
56 / 2 = 28	0
28 / 2 = 14	0
14 / 2 = 7	0
7 / 2 = 3	1
3 / 2 = 1	1
1 / 2 = 0	1

Write remainder from bottom to up (in reverse order), this will be 1110000 which is equivalent binary number of decimal integer 112.

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Number System Conversion

- ✓ **Base conversion from Binary** to octal / hex
  - Binary: 10011110001
  - Octal: 10 | 011 | 110 | 001 = 2361_8
  - Hex: 100 | 1111 | 0001 = $4F1_{16}$
- ✓ Conversion to decimal
  - $101_2 = 1\times2^2 + 0\times2^1 + 1\times2^0 = 5_{10}$

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Number System Conversion

- Octal is the base 8 number system, that only uses the digits 0 through 7.
- Its main advantage is the ease of conversion with binary (base 2), as each digit in octal can be written as a unique three-digit binary number.
- **Converting octal to decimal can be done with repeated division**.

✓ Start the decimal result at 0.

✓ Remove the most significant octal digit (leftmost) and add it to the result.

✓ If all octal digits have been removed, you're done. Stop.

✓ Otherwise, multiply the result by 8.

✓ Go to step 2.

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Octal to Decimal Conversion

Octal Digits	Operation	Decimal Result	Operation	Decimal Result
345	+3	3	× 8	24
45	+4	28	× 8	224
5	+5	229	done.	

- The conversion can also be performed in the conventional mathematical way, by showing each digit place as an increasing power of 8.

- 345 octal = $(3 * 8^2) + (4 * 8^1) + (5 * 8^0) = (3 * 64) + (4 * 8) + (5 * 1) = 229$ decimal

References:-

Video link:

https://nptel.ac.in/courses/108/105/108105113/

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Octal to Hexadecimal Conversion

➢ **Convert 536 from octal to hexadecimal number**

• Convert 536(octal) into its binary equivalent we get $(536)_8$ = (101) (011) (110) =$(101011110)_2$

• Now forming the group of 4 binary bits to obtain its hexadecimal equivalent, $(101011110)_2$= (0001) (0101) (1110) = $(15E)_{16}$

• So the hexadecimal number of 536 is 15E.

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Hexadecimal Code and their Conversion

- ✓ In mathematics and computing, hexadecimal (also base 16, or **hex**) is a positional numeral **system** with a radix, or base, of 16.

- ✓ It uses sixteen distinct symbols -:
  - Symbols 0–9 to represent values zero to nine
  - A, B, C, D, E, F to represent values ten to fifteen.

Hexadecimal to Binary

• Converting from hexadecimal to binary is as easy as converting from binary to hexadecimal. Simply look up each hexadecimal digit to obtain the equivalent group of four binary digits.

Hexadecimal:	0	1	2	3	4	5	6	7
Binary:	0000	0001	0010	0011	0100	0101	0110	0111

Hexadecimal:	8	9	A	B	C	D	E	F
Binary:	1000	1001	1010	1011	1100	1101	1110	1111

• Hexadecimal = A 2 D E

• Binary = 1010 0010 1101 1110 = 1010001011011110 binary

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Hexadecimal to Octal Conversion

- First convert the hexadecimal number into binary and then from binary into octal

.

- For example, to convert A2DE hex into octal:

Hexadecimal =	**A**	**2**	**D**	**E**	
Binary =	**1010**	**0010**	**1101**	**1110**	**= 1010001011011110 binary**

Binary:	000	001	010	011	100	101	110	111
Octal:	0	1	2	3	4	5	6	7

Binary =	001	010	001	011	011	110	
Octal =	1	2	1	3	3	6	= 121336 octal

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Hexadecimal to Decimal

• Converting hexadecimal to decimal can be performed in the conventional mathematical way, by showing each digit place as an increasing power of 16.

• Hexadecimal letter values need to be converted to decimal values before performing the math.

- Hexadecimal: 0 1 2 3 4 5 6 7
- Decimal: 0 1 2 3 4 5 6 7
- Hexadecimal: 8 9 A B C D E F
- Decimal: 8 9 10 11 12 13 14 15

• A2DE hexadecimal:

$= ((A) * 16^3) + (2 * 16^2) + ((D) * 16^1) + ((E) * 16^0)$

$= (10 * 16^3) + (2 * 16^2) + (13 * 16^1) + (14 * 16^0)$

$= (10 * 4096) + (2 * 256) + (13 * 16) + (14 * 1)$

$= 40960 + 512 + 208 + 14$

= 41694 decimal

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Logic Gates



Logic Gates:

- The building blocks used to create digital circuits are **logic gates.**
- There are three elementary logic gates and a range of other simple gates.
- Each gate has its own **logic symbol** which allows complex functions to be represented by a logic diagram.
- The function of each gate can be represented by a **truth table** or using **Boolean notations.**

AND Gate :



(a) Circuit symbol

A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

(b) Truth table

$$C = A \cdot B$$

(c) Boolean expression

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Logic Gates: Basic Gates

- **The OR gate**



(a) Circuit symbol

A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

(b) Truth table

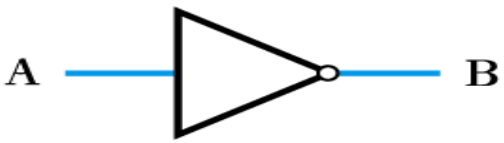
$$C = A + B$$

(c) Boolean expression

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Logic Gates: Basic Gates

- **NOT gate**



(a) Circuit symbol

A	B
0	1
1	0

(b) Truth table

$$B = \overline{A}$$

(c) Boolean expression

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Logic Gates: Universal Gate

NAND gate



(a) Circuit symbol

A	B	C
0	0	1
0	1	1
1	0	1
1	1	0

(b) Truth table

$$C = \overline{A \cdot B}$$

(c) Boolean expression

NOR gate



(a) Circuit symbol

A	B	C
0	0	1
0	1	0
1	0	0
1	1	0

(b) Truth table

$$C = \overline{A + B}$$

(c) Boolean expression

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Logic Gates: Exclusive Gates

- **Exclusive OR gate**



(a) Circuit symbol

A	B	C
0	0	0
0	1	1
1	0	1
1	1	0

(b) Truth table

$$C = A \oplus B$$

(c) Boolean expression

Exclusive NOR gate



(a) Circuit symbol

A	B	C
0	0	1
0	1	0
1	0	0
1	1	1

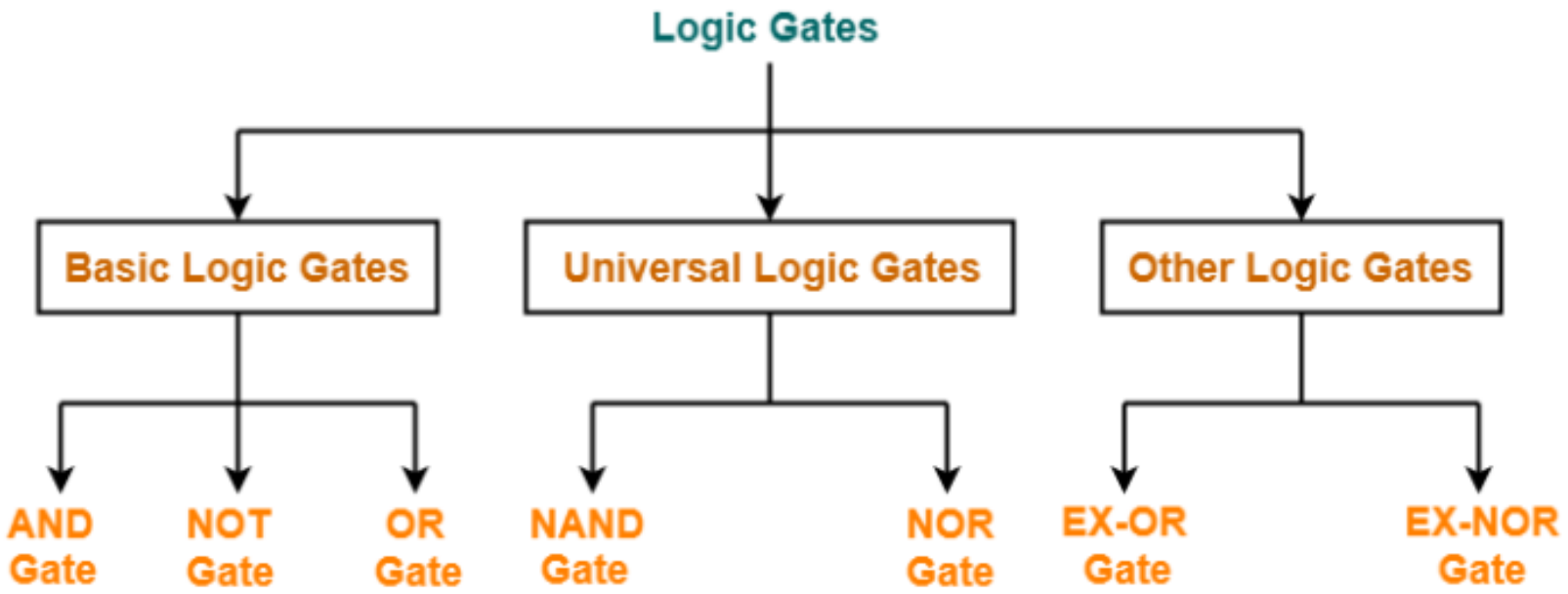
(b) Truth table

$$C = \overline{A \oplus B}$$

(c) Boolean expression

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Types of Logic Gates



Types of Logic Gates

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Adders, Multiplexer and Demultiplexer

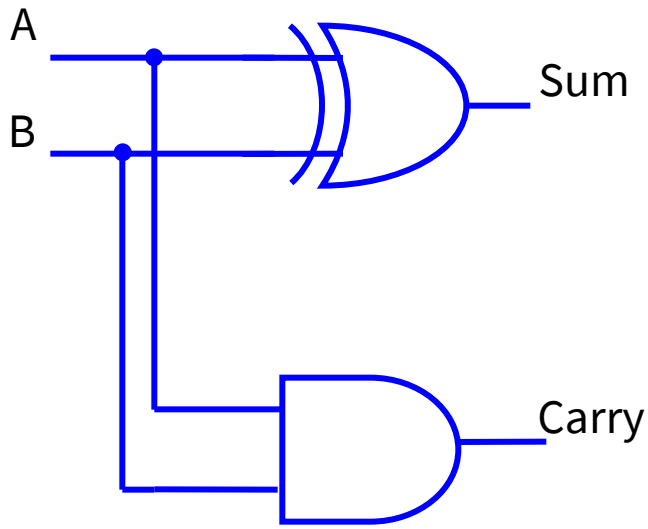


Half Adder Circuit Design using Basic Gates

Truth Table For Half Adder

A	B	S(um)	C(arry)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Circuit Diagram For Half Adder



$$S = \overline{A}B + A\overline{B} = A \oplus B$$

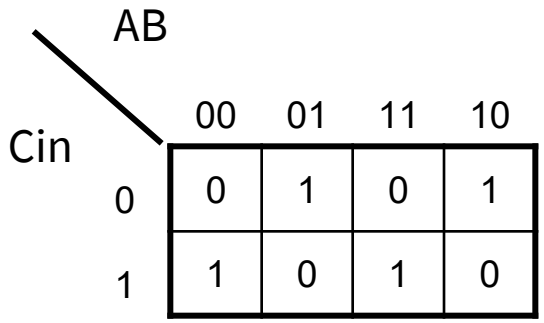
$$C = AB$$

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Full Adder Circuit Design using Basic Gates

Truth Table For Full Adder

Cin	A	B	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



$$S = Cin\overline{A}\,\overline{B} + \overline{Cin}\,\overline{A}B + CinAB + \overline{Cin}A\overline{B}$$

$$= Cin(\overline{A}\,\overline{B} + AB) + \overline{Cin}(\overline{A}B + A\overline{B})$$

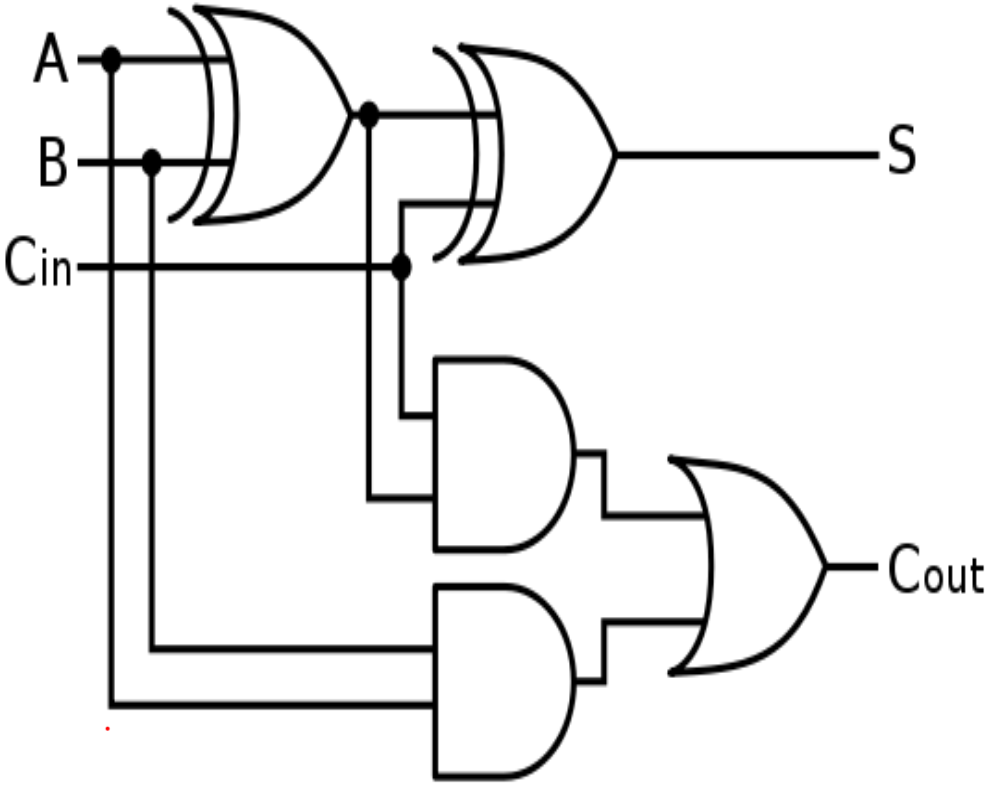
$$= Cin(\overline{A \oplus B}) + \overline{Cin}(A \oplus B)$$

$$= Cin \oplus A \oplus B$$

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Full Adder Circuit Design using Basic Gates

Circuit Diagram For Full Adder



C \ AB	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$$S = Cin \oplus A \oplus B$$

$$Cout = AB + Cin(A \oplus B)$$

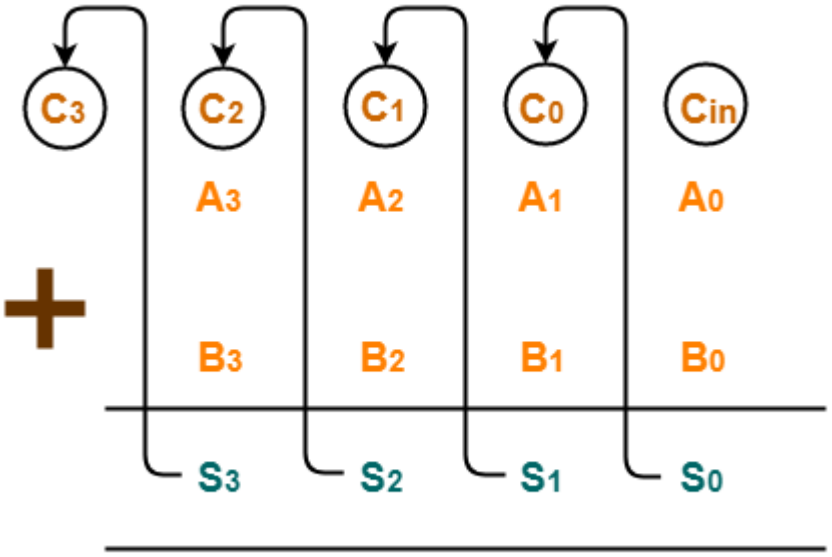
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Ripple Carry Adder

- It is used for the purpose of adding two n-bit binary numbers.
- It requires n full adders in its circuit for adding two n-bit binary numbers.
- It is also known as **n-bit parallel adder**.

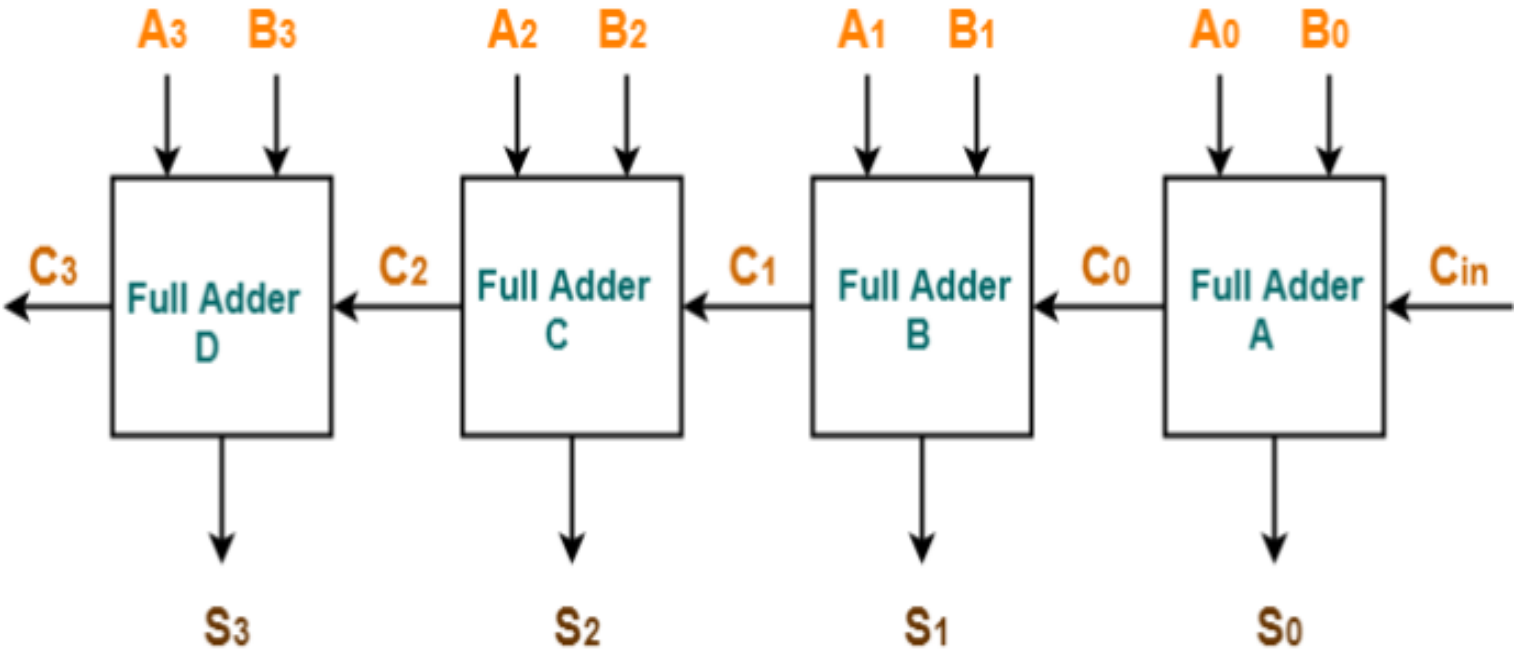
4-bit Ripple Carry Adder-

Any two 4-bit binary numbers $A_3A_2A_1A_0$ and $B_3B_2B_1B_0$ can be added as:



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Using ripple carry adder, this addition is carried out as shown by the following logic diagram-



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Why Ripple Carry Adder is called so?

➢ The carry out produced by each full adder serves as carry-in for its adjacent most significant full adder.

➢ Each carry bit ripples or waves into the next stage.

➢ That's why, it is called as **"Ripple Carry Adder"**.

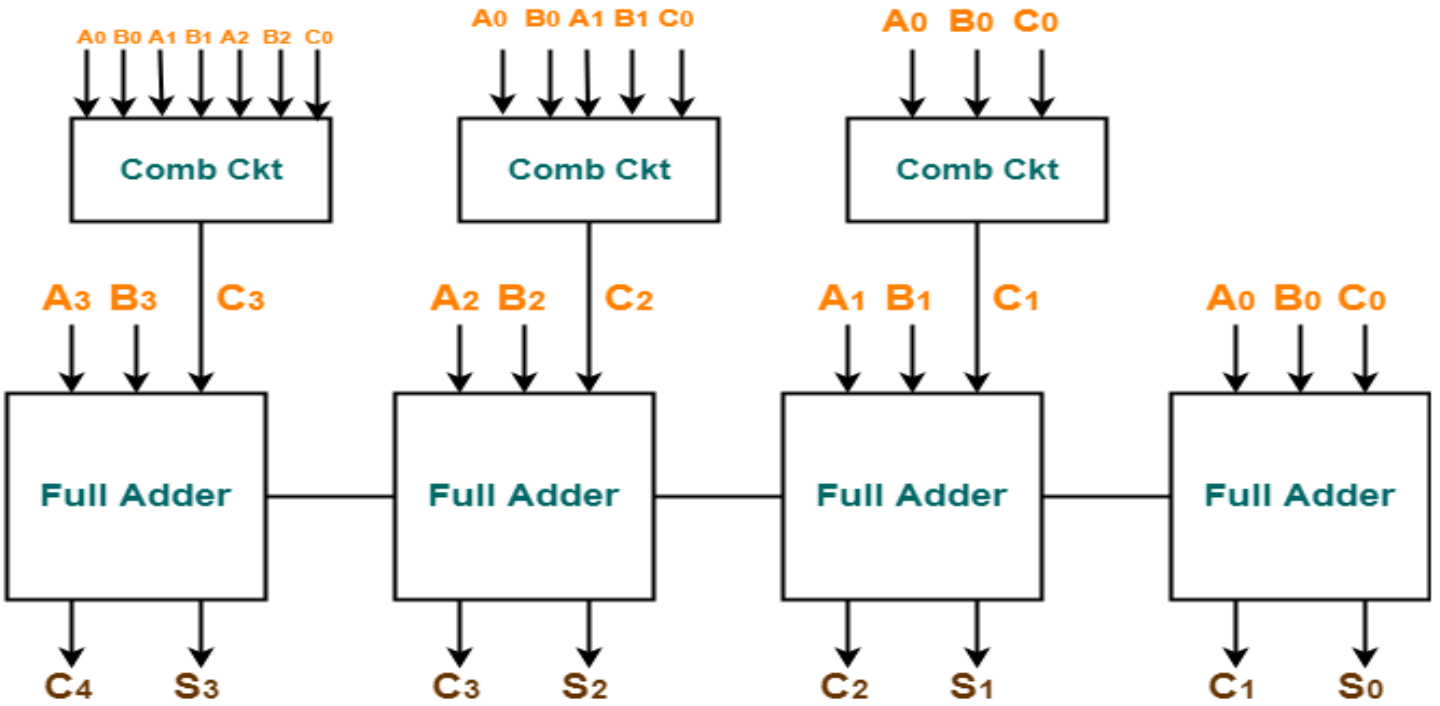
Disadvantages :

- Ripple Carry Adder does not allow to use all the full adders simultaneously.
- Each full adder has to necessarily wait until the carry bit becomes available from its adjacent full adder.
- This increases the propagation time.
- Due to this reason, ripple carry adder becomes extremely slow.

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Carry Look Ahead Adder

- Carry Look Ahead Adder is an improved version of the ripple carry adder.
- It generates the carry-in of each full adder simultaneously without causing any delay.



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Carry Look Ahead Adder

- The carry-in of any stage full adder depends only on the following two parameters-
  - Bits being added in the previous stages
  - Carry-in provided in the beginning

- The above two parameters are always known from the beginning.

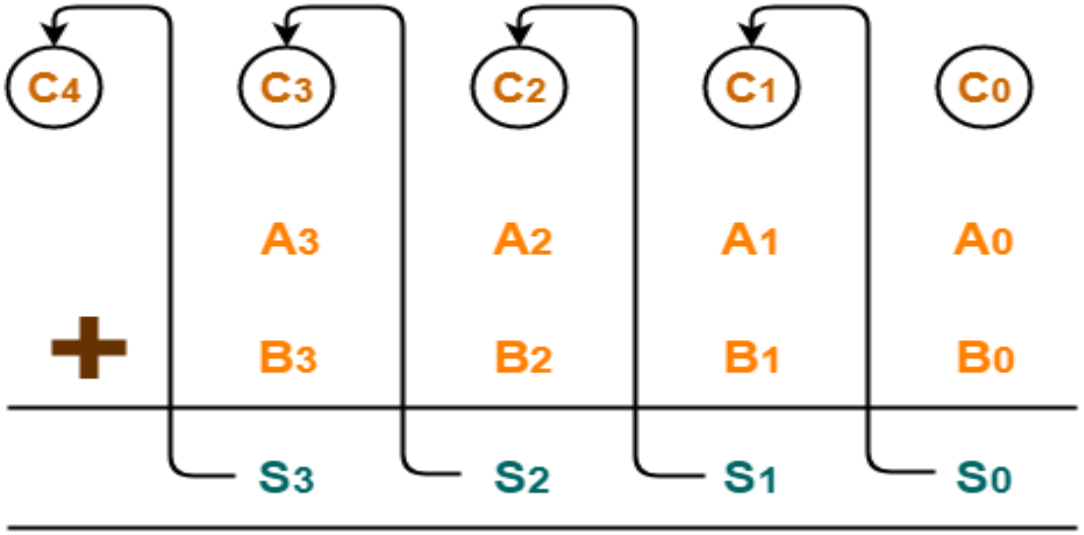
- So, the carry-in of any stage full adder can be evaluated at any instant of time.

- Thus, any full adder need not wait until its carry-in is generated by its previous stage full adder.

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4-Bit Carry Look Ahead Adder-

Suppose two 4-bit binary numbers $A_3A_2A_1A_0$ and $B_3B_2B_1B_0$ are to be added.



$$C_1 = C_0 (A_0 \oplus B_0) + A_0B_0$$
$$C_2 = C_1 (A_1 \oplus B_1) + A_1B_1$$
$$C_3 = C_2 (A_2 \oplus B_2) + A_2B_2$$
$$C_4 = C_3 (A_3 \oplus B_3) + A_3B_3$$

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4-Bit Carry Look Ahead Adder-

For simplicity, Let-

- $G_i = A_iB_i$ where G is called carry generator
- $P_i = A_i \oplus B_i$ where P is called carry propagator

Then, re-writing the above equations,

$C_1 = C_0P_0 + G_0$ …….. (1)

$C_2 = C_1P_1 + G_1$ …….. (2)

$C_3 = C_2P_2 + G_2$ …….. (3)

$C_4 = C_3P_3 + G_3$ …….. (4)

- C1, C2 and C3 are intermediate carry bits.
- So, let's remove C_1, C_2 and C_3 from RHS of every equation.
- Substituting (1) in (2), we get C_2 in terms of C_0.
- substituting (2) in (3), we get C_3 in terms of C_0 and so on.

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4-Bit Carry Look Ahead Adder-

Finally, we have the following equations-

$C_1 = C_0P_0 + G_0$
$C_2 = C_0P_0P_1 + G_0P_1 + G_1$
$C_3 = C_0P_0P_1P_2 + G_0P_1P_2 + G_1P_2 + G_2$
$C_4 = C_0P_0P_1P_2P3 + G_0P_1P_2P_3 + G_1P_2P_3 + G_2P_3 + G_3$

➢ These equations show that the carry-in of any stage full adder depends only on-

- Bits being added in the previous stages
- Carry bit which was provided in the beginning

Advantages of Carry Look Ahead Adder-

• It generates the carry-in for each full adder simultaneously.

• It reduces the propagation delay.

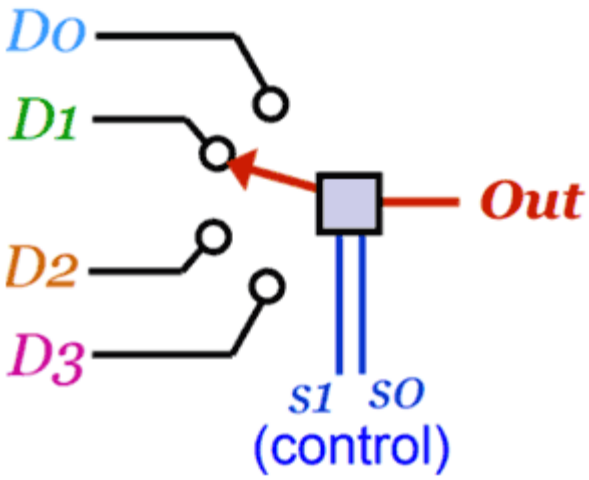
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Multiplexer

➢A plain example of a mechanical Multiplexer.

➢ In electronic circuit which involves high speed switching and data transfers we should be able to select the required input very fast using digital circuits

➢The Control signals (S1 and S0) does exactly the same, they select one input of the many available ones based on the signal provided to them.



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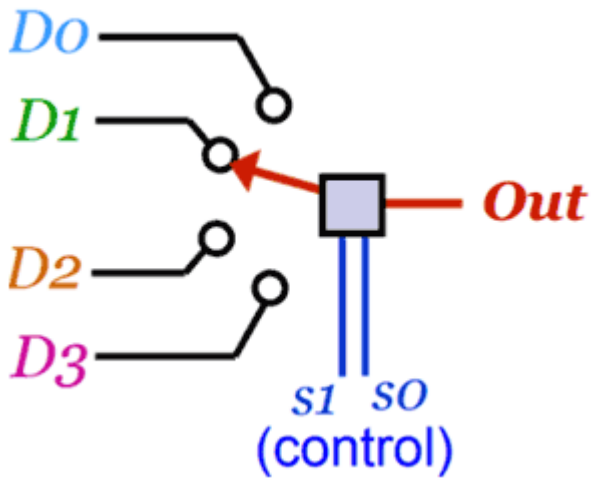
Multiplexer

• **Input Pins:** These are the available signal pins from which one has to be selected.

• **Output Pin:** A multiplexer will always have only one output pin. The selected input pin signal will be provided by the output pin.

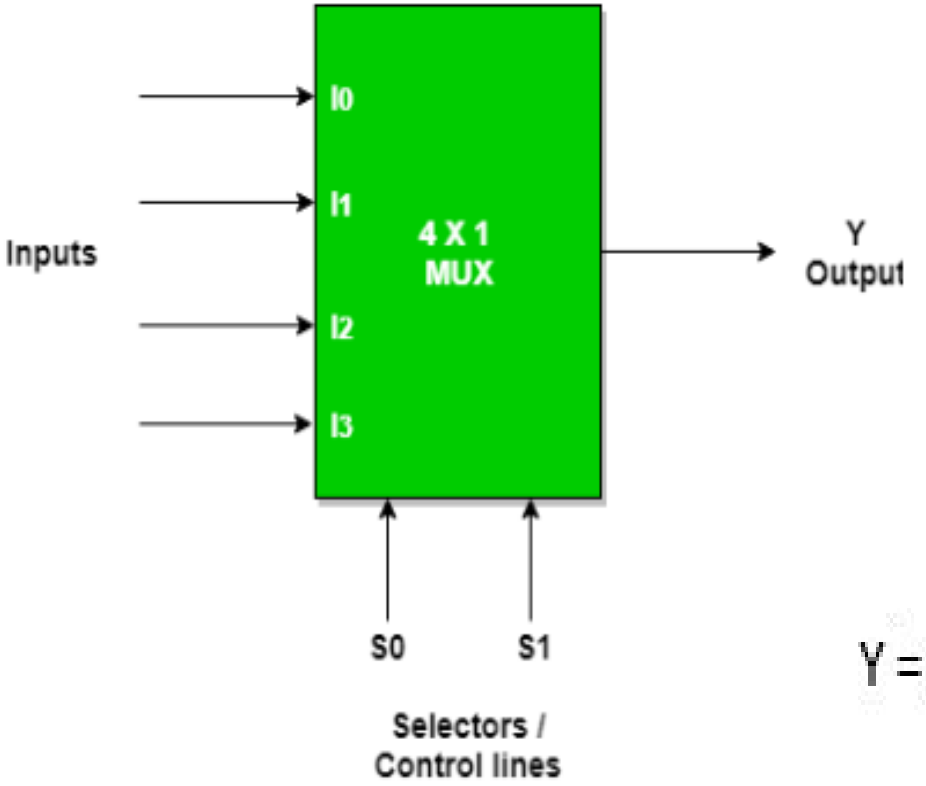
• **Control Pin:** The Control Pins are used to select the input pin signal. The number of Control pins on a Multiplexer depends on the number of input pins.

S1	S0	Out
0	0	D0
0	1	D1
1	0	D2
1	1	D3



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4:1 Multiplexer



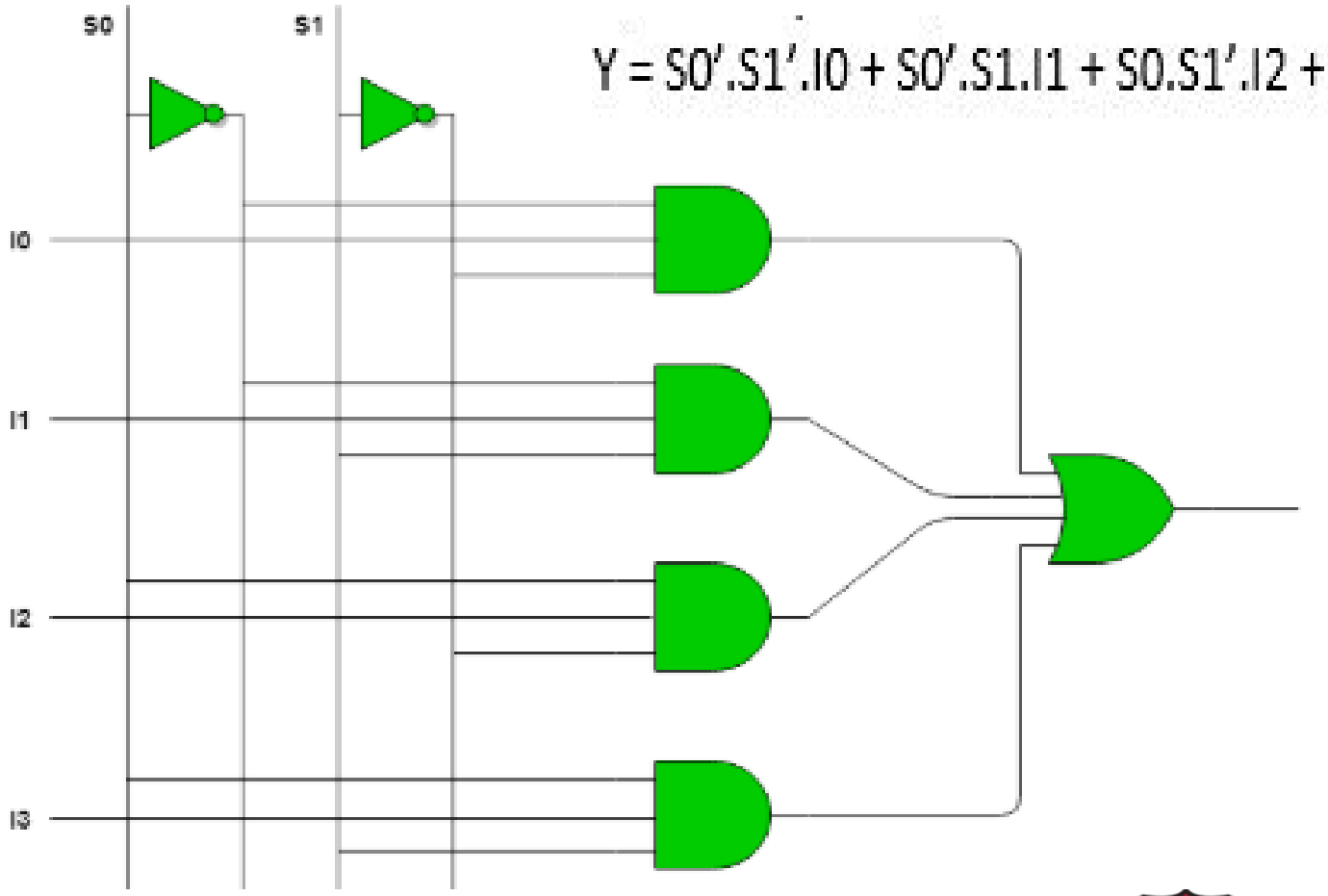
Truth Table

S0	S1	Y
0	0	I0
0	1	I1
1	0	I2
1	1	I3

$$Y = S0'.S1'.I0 + S0'.S1.I1 + S0.S1'.I2 + S0.S1.I3$$

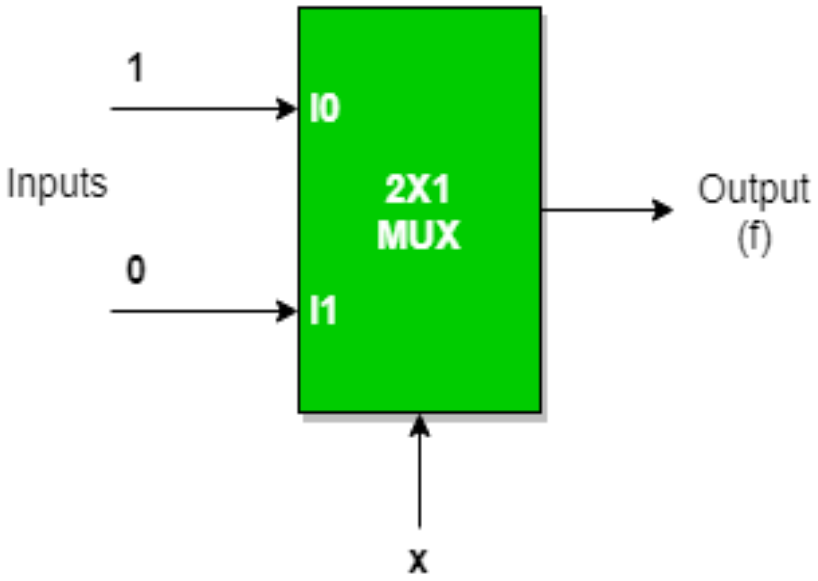
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4:1 Multiplexer



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Implementation of NOT gate using 2 : 1 Mux



Truth Table

x	f
0	1
1	0

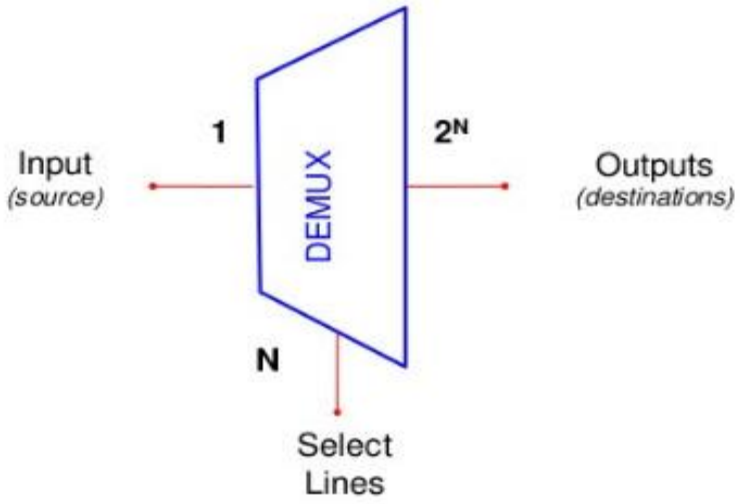
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The Demultiplexer

•The demultiplexer is a combinational logic circuit designed to switch one common input line to one of several seperate output line

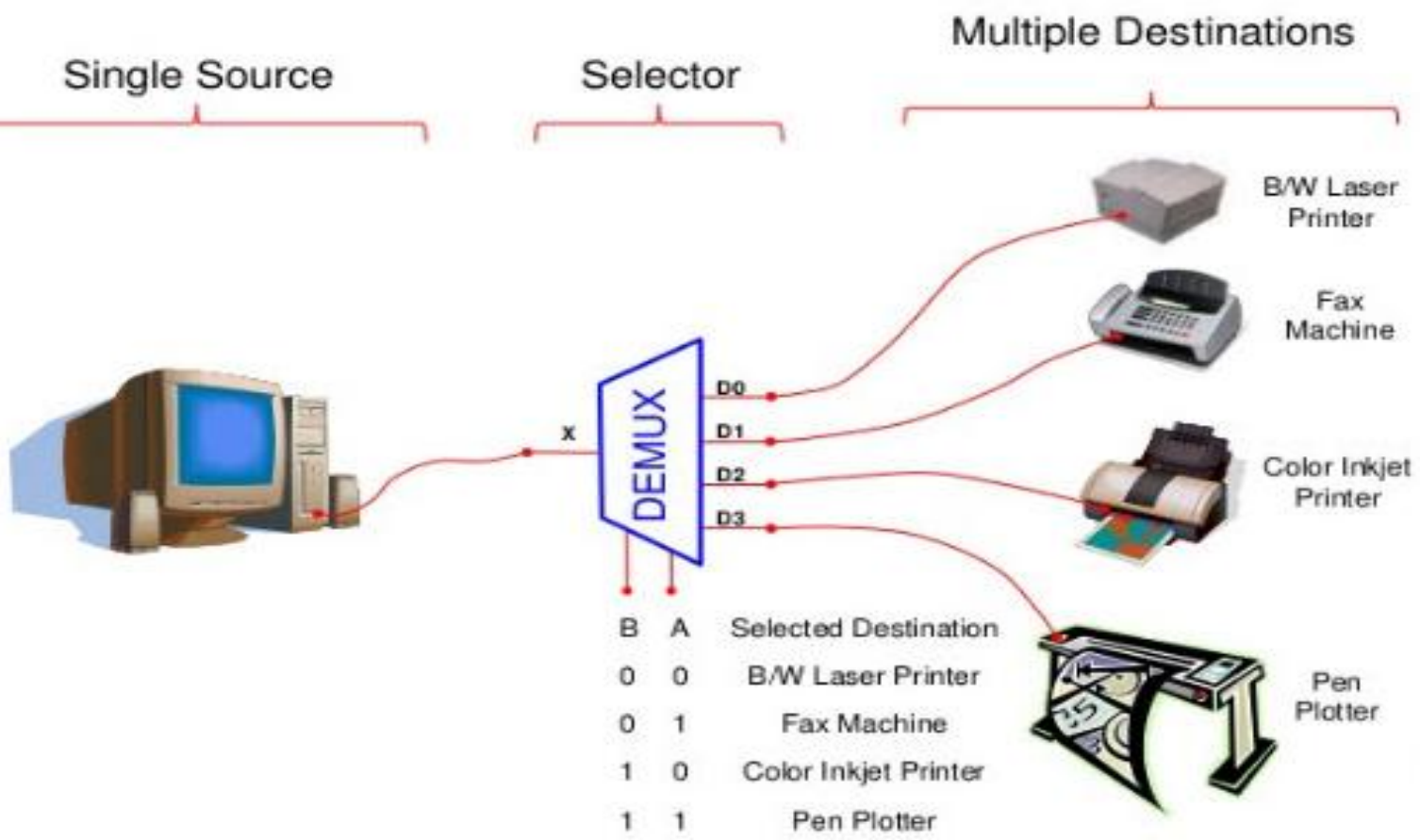
•The data distributor, known more commonly as a **Demultiplexer** or “Demux” for short, is the exact opposite of the Multiplexer

•The *demultiplexer* takes one single input data line and then switches it to any one of a number of individual output lines one at a time.



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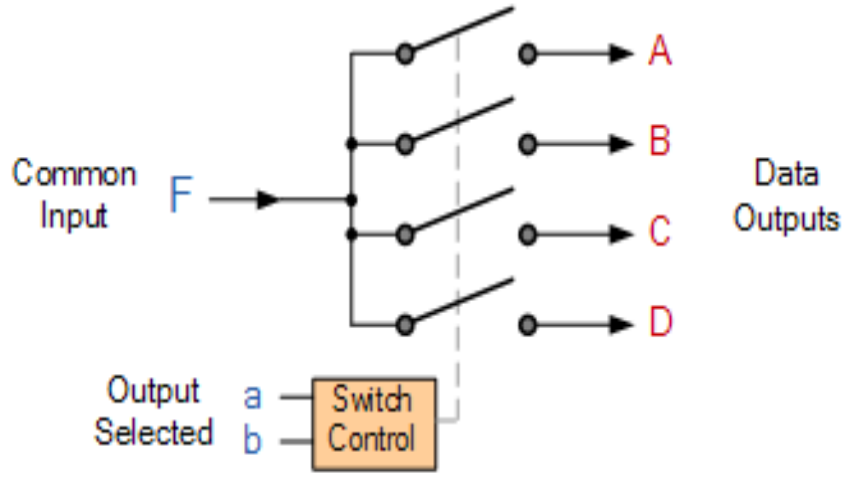
Demultiplexer



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1-to-4 Channel De-multiplexer

The function of the **Demultiplexer** is to switch one common data input line to any one of the 4 output data lines A to D

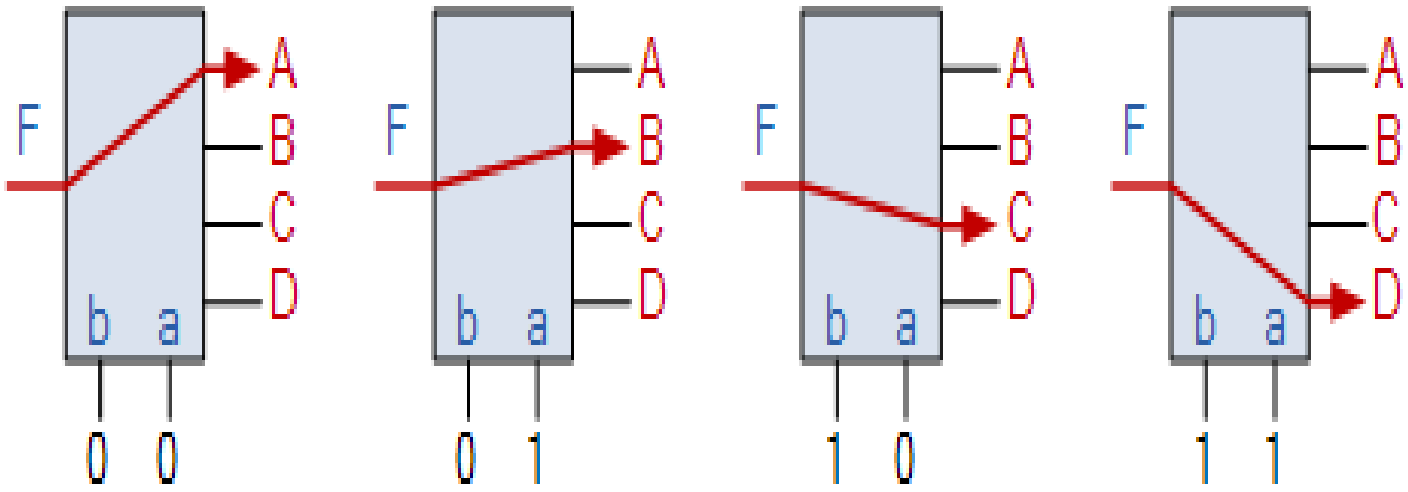


Output Select		Data Output Selected
a	b	
0	0	A
0	1	B
1	0	C
1	1	D

F = a'b'A + a'bB + ab'C + abD

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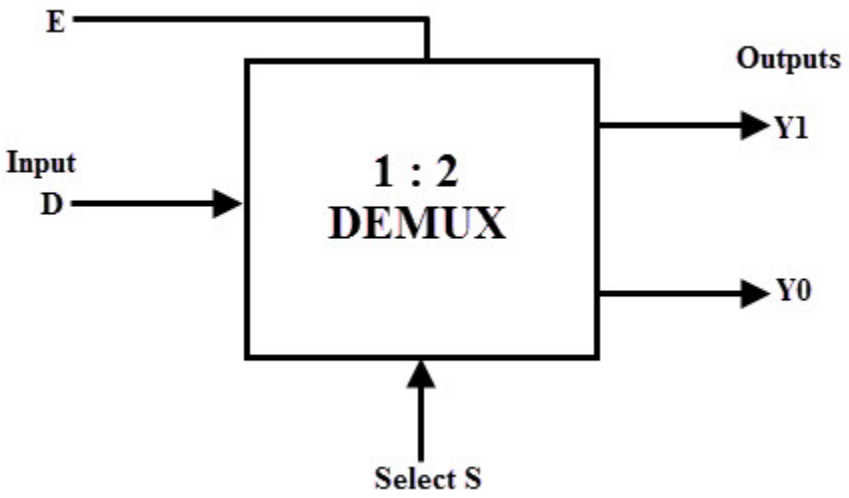
Demultiplexer Output Line Selection



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1-to-2 Demultiplexer

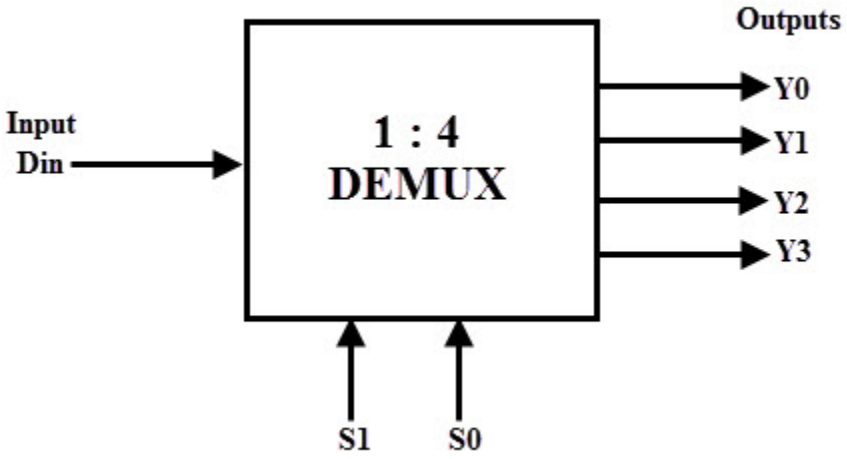
•A 1-to-2 demultiplexer consists of one input line, two output lines and one select line.
• The signal on the select line helps to switch the input to one of the two outputs.



Select	Input	Outputs	
S	D	Y_1	Y_0
0	0	0	0
0	1	0	1
1	0	0	0
1	1	1	0

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1:4 Demux



Data Input	Select Inputs		Outputs			
D	S_1	S_0	Y_3	Y_2	Y_1	Y_0
D	0	0	0	0	0	D
D	0	1	0	0	D	0
D	1	0	0	D	0	0
D	1	1	D	0	0	0

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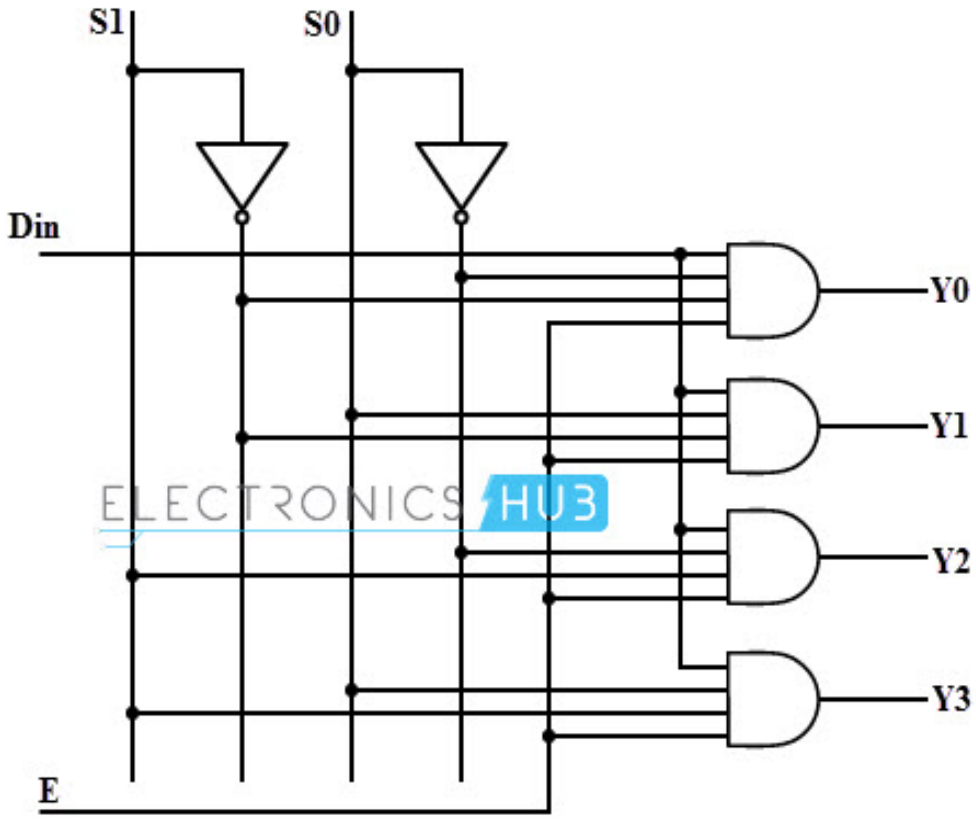
1:4 Demux using Gates

$$Y0 = \overline{S1}\ \overline{S0}\ D$$

$$Y1 = \overline{S1}\ S0\ D$$

$$Y2 = S1\ \overline{S0}\ D$$

$$Y3 = S1\ S0\ D$$



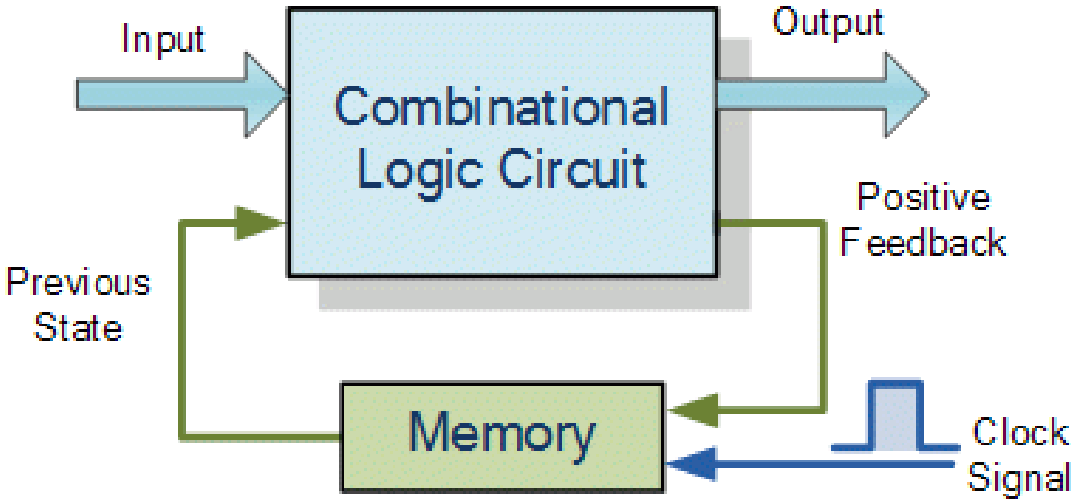
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Flip-Flop



Sequential Circuit: Introduction

Unlike **Combinational Logic** circuits that change state depending upon the actual signals being applied to their inputs at that time, **Sequential Logic** circuits have some form of inherent “Memory” built in.



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Sequential Circuit: Introduction

- Most digital systems like digital watches, digital phones, digital computers, digital traffic light controllers and so on require memory elements.

- Memory elements are digital circuits that can store and retrieve data in the form of 1's and 0's.

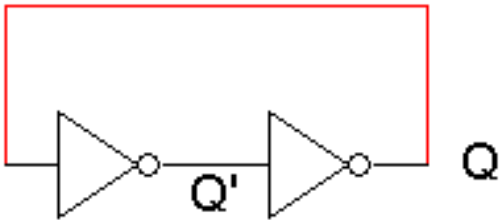
- The output of the systems with memory depends not only on present inputs but also on what has happened in the past.

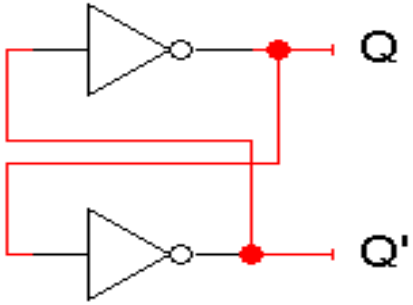
- SR latch is an example of memory circuits that can store one bit of information

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One Bit Memory Cell

How can a circuit “remember” anything, when it’s just a bunch of gates that produce outputs according to the inputs?
The basic idea is to make a loop, so the circuit outputs are also inputs.
Here is one initial attempt, shown with two equivalent layouts:





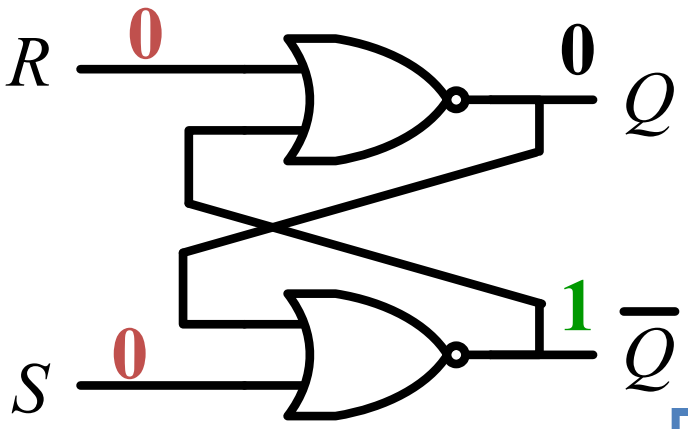
Does this satisfy the properties of memory?

These circuits “remember” Q, because its value never changes. (Similarly, Q’ never changes either.)
We can also “read” Q, by attaching a probe or another circuit.
But we can’t *change* Q! There are no external inputs here, so we can’t control whether Q=1 or Q=0.

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Latches: *SR* Latch

A	B	Nor
0	0	1
0	1	0
1	0	0
1	1	0

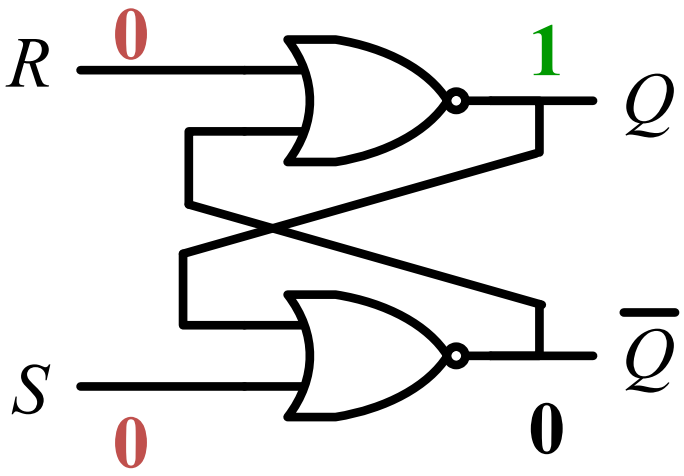


S	R	Q_0	Q	Q'
0	0	0		

Initial Value

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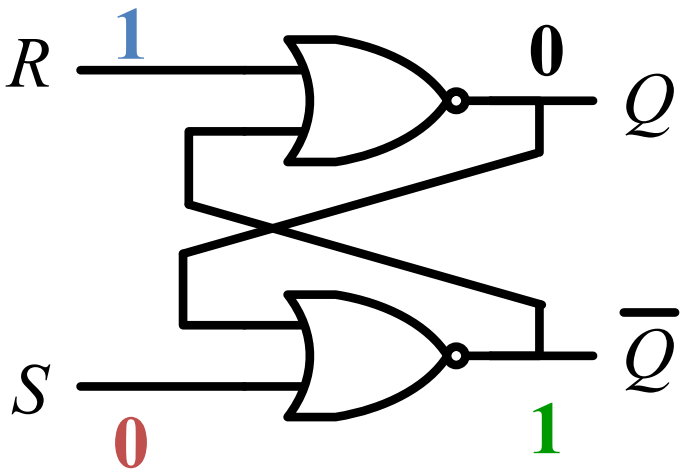
Latches: *SR* Latch



S	R	Q_0	Q	Q'	
0	0	0	0	1	$Q = Q_0$
0	0	1	1	0	$Q = Q_0$

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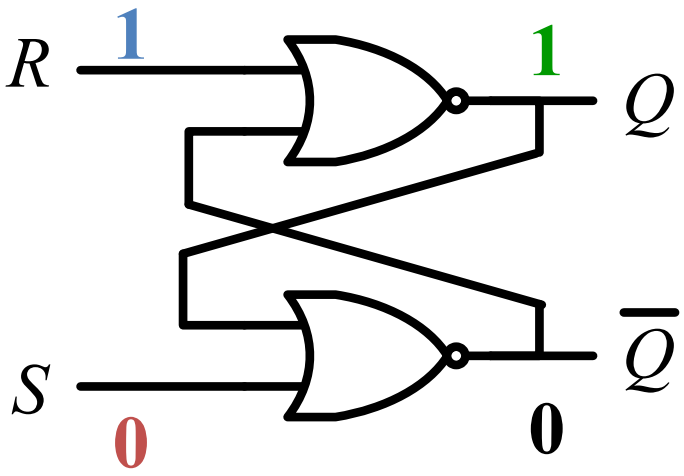
Latches: *SR* Latch



S	R	Q_0	Q	Q'	
0	0	0	0	1	$Q = Q_0$
0	0	1	1	0	
0	1	0	0	1	$Q = 0$

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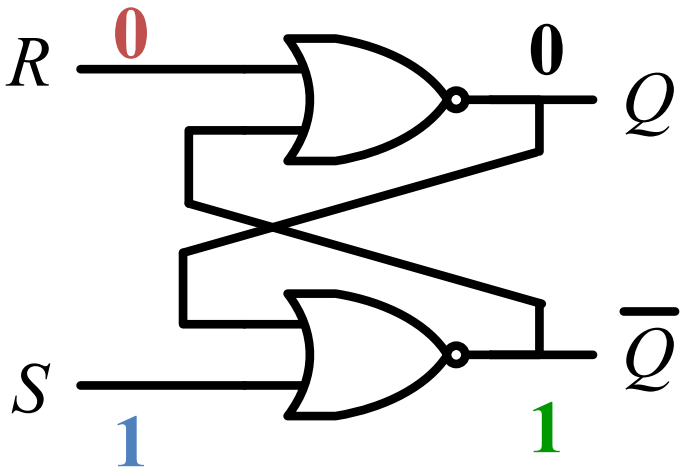
Latches: *SR* Latch



S	R	Q_0	Q	Q'	
0	0	0	0	1	$Q = Q_0$
0	0	1	1	0	
0	1	0	0	1	$Q = 0$
0	1	1	0	1	$Q = 0$

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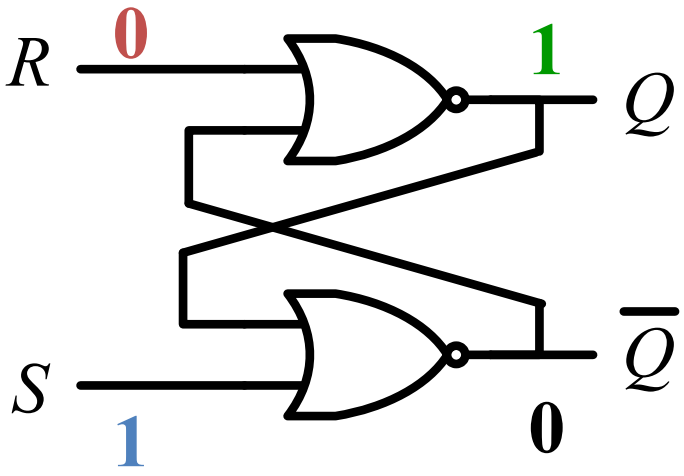
Latches: *SR* Latch



S	R	Q_0	Q	Q'	
0	0	0	0	1	$Q = Q_0$
0	0	1	1	0	
0	1	0	0	1	$Q = 0$
0	1	1	0	1	
1	0	0	1	0	$Q = 1$

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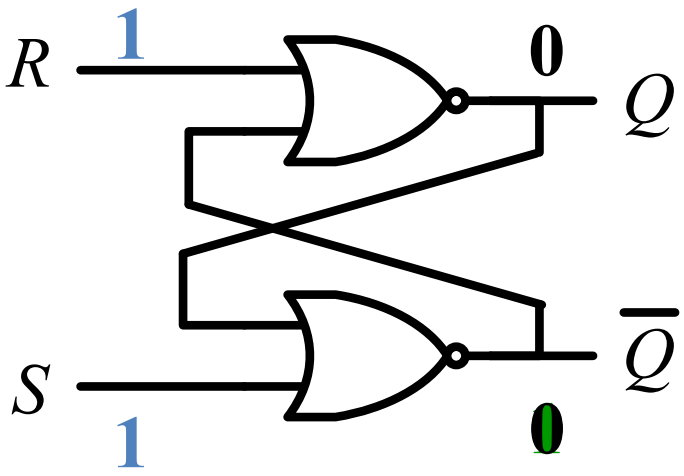
Latches: *SR* Latch



S	R	Q_0	Q	Q'	
0	0	0	0	1	$Q = Q_0$
0	0	1	1	0	
0	1	0	0	1	$Q = 0$
0	1	1	0	1	
1	0	0	1	0	$Q = 1$
1	0	1	1	0	$Q = 1$

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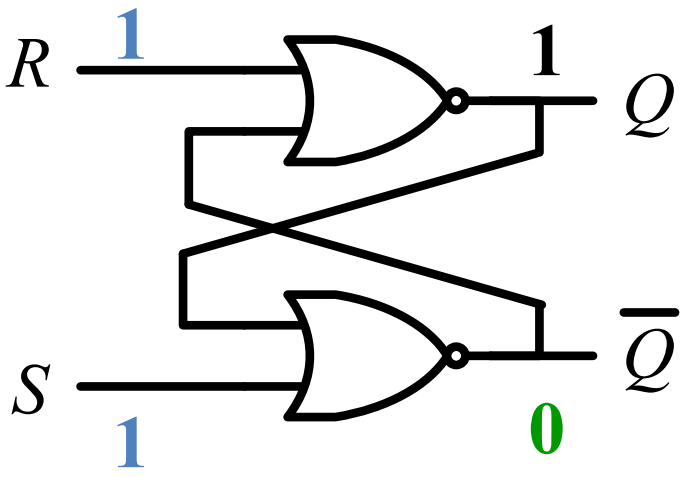
Latches: *SR* Latch



S	R	Q_0	Q	Q'	
0	0	0	0	1	$Q = Q_0$
0	0	1	1	0	
0	1	0	0	1	$Q = 0$
0	1	1	0	1	
1	0	0	1	0	$Q = 1$
1	0	1	1	0	
1	1	0	0	0	$Q = Q'$

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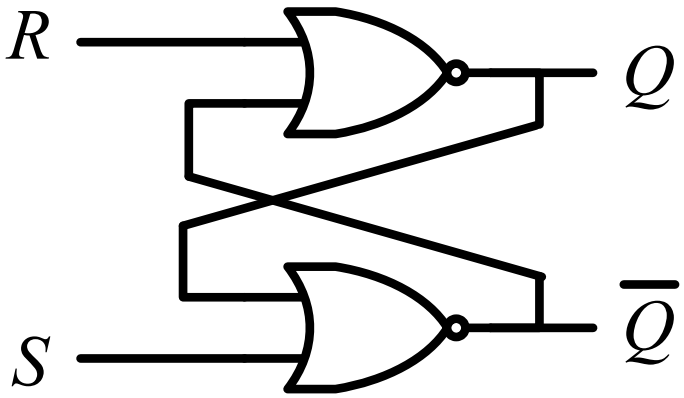
Latches: *SR* Latch



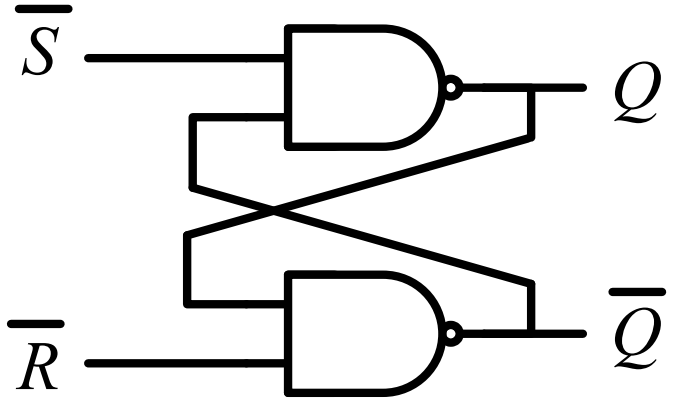
S	R	Q_0	Q	Q'	
0	0	0	0	1	$Q = Q_0$
0	0	1	1	0	
0	1	0	0	1	$Q = 0$
0	1	1	0	1	
1	0	0	1	0	$Q = 1$
1	0	1	1	0	
1	1	0	0	0	$Q = Q'$
1	1	1	0	0	

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Latches: *SR* Latch



S	R	Q	
0	0	Q_0	No change
0	1	0	Reset
1	0	1	Set
1	1	Q=Q'=0	Invalid



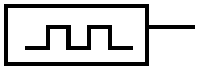
S'	R'	Q	
0	0	Q=Q'=1	Invalid
0	1	1	Set
1	0	0	Reset
1	1	Q_0	No change

Clocks and synchronization

- A clock is a special device that whose output continuously alternates between 0 and 1.

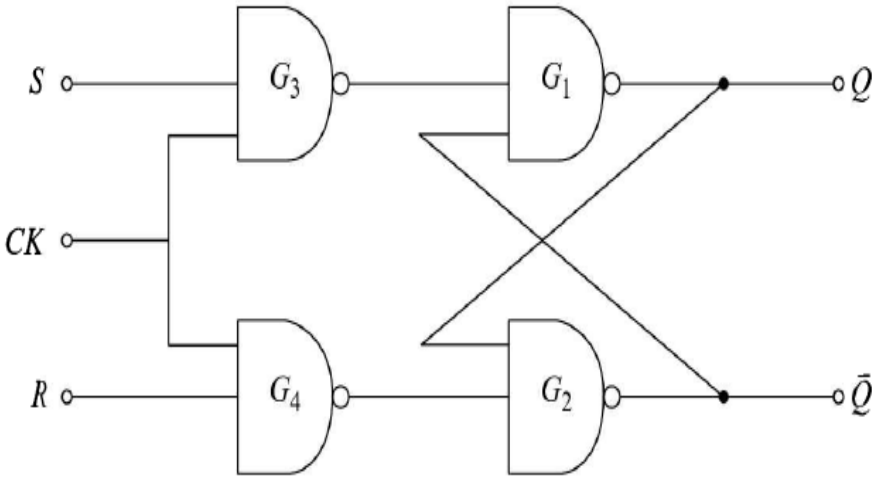
clock period



- The time it takes the clock to change from 1 to 0 and b  led the clock period, or clock cycle time.
- The clock frequency is the inverse of the clock period. The unit of measurement for frequency is the hertz.
- Clocks are often used to synchronize circuits.
  - They generate a repeating, predictable pattern of 0s and 1s that can trigger certain events in a circuit, such as writing to a latch.
  - If several circuits share a common clock signal, they can coordinate their actions with respect to one another.
- This is similar to how humans use real clocks for synchronization.

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Gated or Clocked SR Flip-Flop



- This extra conditional input is called an “Enable” input and is given the prefix of “EN“.
- The addition of this input means that the output at Q only changes state when it is HIGH and can therefore be used as a clock (CLK) input making it level-sensitive.

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Gated or Clocked SR Flip-Flop

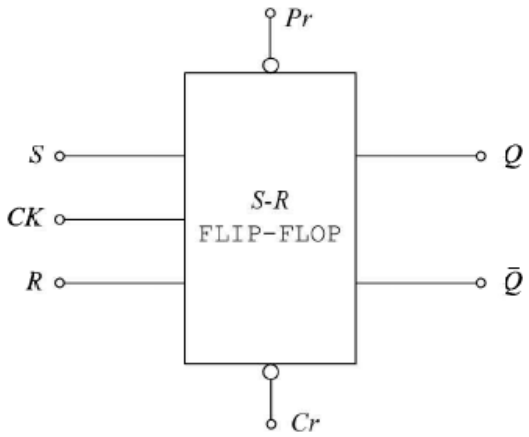
➢**Gated Bistable SR Flip-flop** operates as a standard bistable latch but the outputs are only activated when a logic “1” is applied to its EN input and deactivated by a logic “0”.

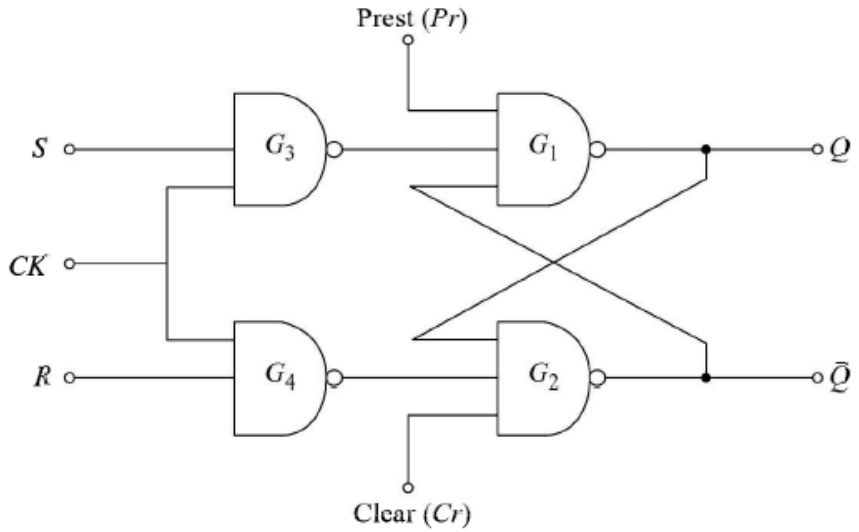
C	S	R	S′	R′	Q
0	x	x	1	1	No change
1	0	0	1	1	No change
1	0	1	1	0	0 (reset)
1	1	0	0	1	1 (set)
1	1	1	0	0	Avoid!

Truth Table-SR Flip Flop

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SR Flip-Flop with Preset and Clear





Inputs			Output	Operation performed
CK	*Cr*	*Pr*	*Q*	
1	1	1	Q_{n+1}(Table 7.1)	Normal FLIP-FLOP
0	0	1	0	Clear
0	1	0	1	Preset

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SR Flip Flop vs JK Flip Flop

The basic S-R NAND flip-flop circuit has many advantages and uses in sequential logic circuits but it suffers from two basic switching problems.

1. the Set = 0 and Reset = 0 condition (S = R = 0) must always be avoided.

2. if Set or Reset change state while the enable (EN) input is high the correct latching action may not occur.

Then to overcome these two fundamental design problems with the SR flip-flop design, the **JK flip Flop** was developed.

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Master-Slave JK Flip-flop

➢The **Master-Slave Flip-Flop** is basically two gated SR flip-flops connected together in a series configuration with the slave having an inverted clock pulse.

➢The outputs from Q and Q' from the "Slave" flip-flop are fed back to the inputs of the "Master" with the outputs of the "Master" flip flop being connected to the two inputs of the "Slave" flip flop.

➢This feedback configuration from the slave's output to the master's input gives the characteristic toggle of the JK flip flop.

➢The input signals J and K are connected to the gated "master" SR flip flop which "locks" the input condition while the clock (Clk) input is "HIGH" at logic level "1".

➢ As the clock input of the "slave" flip flop is the inverse (complement) of the "master" clock input, the "slave" SR flip flop does not toggle.

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Master-Slave JK Flip-flop

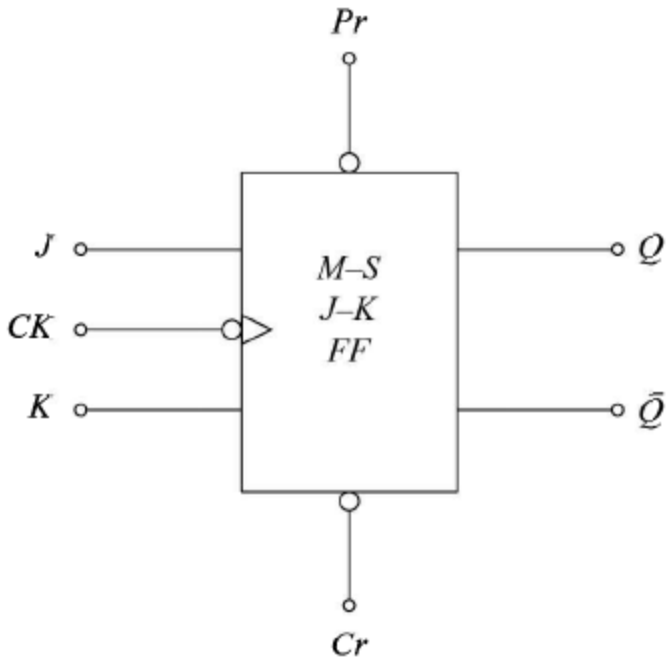
Then on the “Low-to-High” transition of the clock pulse the inputs of the “master” flip flop are fed through to the gated inputs of the “slave” flip flop and on the “High-to-Low” transition the same inputs are reflected on the output of the “slave” making this type of flip flop edge or pulse-triggered.

The circuit accepts input data when the clock signal is “HIGH”, and passes the data to the output on the falling-edge of the clock signal.

The **Master-Slave JK Flip flop** is a “Synchronous” device as it only passes data with the timing of the clock signal.

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Master-Slave JK Flip-flop Logic Symbol



At the clock input terminal > indicates that output changes when clock makes a transition and bubble indicates the negative transition.

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Characteristic equation of J-K Flip Flop

- We can also write characteristic equations, where the next state Q(t+1) is defined in terms of the current state Q(t) and inputs.

Truth Table:

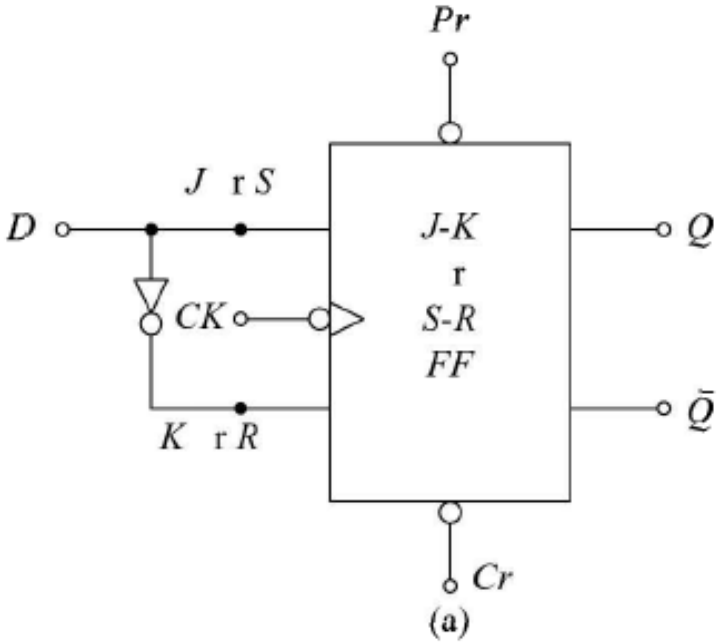
J	K	Q(t+1)	Operation
0	0	Q(t)	No change
0	1	0	Reset
1	0	1	Set
1	1	Q'(t)	Complement

Characteristic Equation:

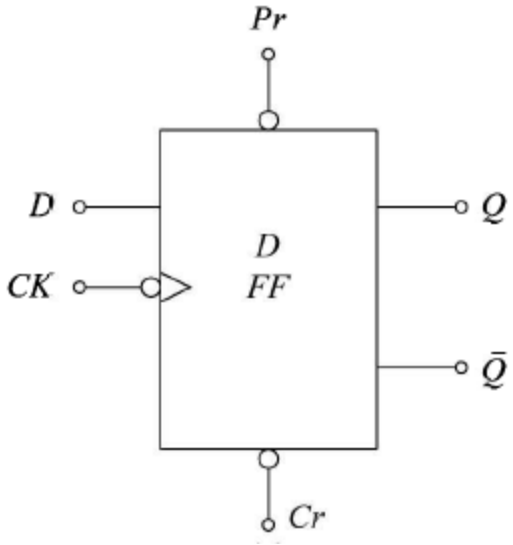
$$Q(t+1) = K'Q(t) + JQ'(t)$$

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D-Type Flip-flop



JK/SR Flip Flop converted into D Flip Flop



D Flip Flop Logic Symbol

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D-Type Flip-flop

- The “D flip flop” will store and output whatever logic level is applied to its data terminal so long as the clock input is HIGH.
- Once the clock input goes LOW the “set” and “reset” inputs of the flip-flop are both held at logic level “1” so it will not change state and store whatever data was present on its output before the clock transition occurred.
- In other words the output is “latched” at either logic “0” or logic “1”.

Truth Table:

D	Q(t+1)	Operation
0	0	Reset
1	1	Set

Characteristic Equation:

$$Q(t+1) = D$$

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References

Reading Material

- https://www.cs.princeton.edu/courses/archive/spr15/cos217/lectures/03_NumberSystems.pdf
- https://www.site.uottawa.ca/~petriu/Digital-Logic.pdf
- https://www.cs.ucr.edu/~ehwang/courses/cs120b/flipflops.pdf

Video Link

- https://youtu.be/CeD2L6KbtVM
- https://youtu.be/yLP0vFSbCLg
- https://www.youtube.com/watch?v=jm0PGDSSBkI

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Thank You

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Analog to Digital Signal Conversion

Sampling :This is measuring the analog signal's value at regular intervals.
Quantization: Converts the continuous amplitude values into discrete levels.The number of levels depends on the **bit resolution**.
For example, a 4-bit ADC has 2^4=16 quantization levels
Encoding: Each quantized level is assigned a binary number.

- Total Levels = $2^4 = 16$

So each level =

$$\text{Step Size} = \frac{5V - 0V}{16 - 1} = \frac{5}{15} \approx 0.333V$$

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Voltage Range (V)	Digital Output (Binary)
0.00 – 0.33	0000
0.34 – 0.66	0001
0.67 – 0.99	0010
1.00 – 1.32	0011
1.33 – 1.65	0100
1.66 – 1.98	0101
1.99 – 2.31	0110
2.32 – 2.64	0111
2.65 – 2.97	1000
2.98 – 3.30	1001
3.31 – 3.63	1010
3.64 – 3.96	1011
3.97 – 4.29	1100
4.30 – 4.62	1101
4.63 – 4.95	1110
4.96 – 5.00	1111

If I want to find out what is digital equivalent of 2.4 V?

2.32 – 2.64 0111

Output will be 0111

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